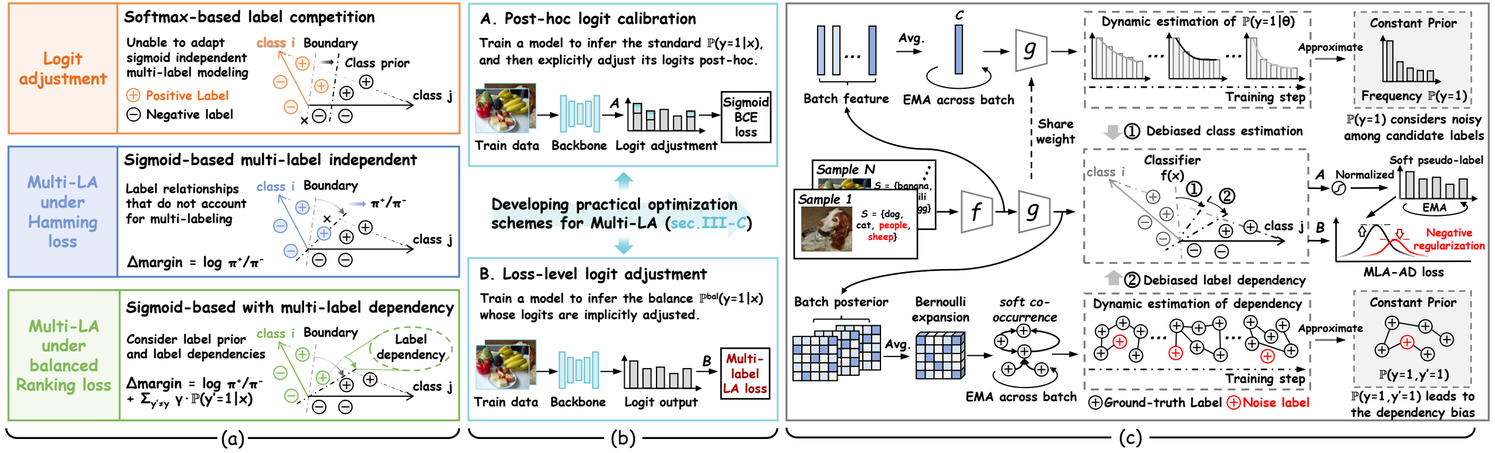

Logit adjustment
Softmax-based label competition
Unable to adapt sigmoid independent multi-label modeling
⊕ Positive Label
⊖ Negative label
class i
Boundary
Class prior
class j
Multi-LA under Hamming loss
Sigmoid-based multi-label independent
Label relationships that do not account for multi-labeling
π⁺/π⁻
Δmargin = log π⁺/π⁻
Multi-LA under balanced Ranking loss
Sigmoid-based with multi-label dependency
Consider label prior and label dependencies
Label dependency
Δmargin = log π⁺/π⁻ + Σ_{y'≠y} γ · ℙ(y'=1|x)
(a)
A. Post-hoc logit calibration
Train a model to infer the standard ℙ(y=1|x), and then explicitly adjust its logits post-hoc.
Train data
Backbone
Logit adjustment
A
Sigmoid BCE loss
Developing practical optimization schemes for Multi-LA (sec.III-C)
B. Loss-level logit adjustment
Train a model to infer the balance ℙ^bal(y=1|x) whose logits are implicitly adjusted.
Logit output
B
Multi-label LA loss
(b)
Batch feature
Avg.
EMA across batch
C
g
Share weight
Dynamic estimation of ℙ(y=1|θ)
Training step
Approximate
Constant Prior
Frequency ℙ(y=1)
ℙ(y=1) considers noisy among candidate labels
① Debiased class estimation
Sample N
S = {banana, ...}
Sample 1
S = {dog, cat, people, sheep}
f
Classifier f(x)
Soft pseudo-label
Normalized
EMA
Negative regularization
MLA-AD loss
② Debiased label dependency
Batch posterior
Bernoulli expansion
soft co-occurrence
Dynamic estimation of dependency
ℙ(y=1,y'=1)
ℙ(y=1,y'=1) leads to the dependency bias
⊕Ground-truth Label ⊕Noise label
(c)